REQUIREMENT ANALYSIS

Solution Requirements • Functional & Non-functional

Student	Harshvardhan Gupta	**Course**	BCA — Computer Applications
Year / Semester	Second Year / 3rd Semester	**College**	MEDI-CAPS University
Enrollment ID	en25ca3020071@medicaps.ac.in	**Project Type**	Individual Project
Project Title	Global Fuel Prices and Taxation Analysis (2020–2026)	**Maximum Marks**	4 marks

1. Functional Requirements

ID	Quality attribute	Requirement / constraint
NFR-1	Usability	Use clear titles, readable labels, consistent units (USD per litre and percentage), and straightforward comparisons. This is a design requirement, not the result of formal usability testing.
NFR-2	Security & Privacy	The workbook is published on Tableau Public and is publicly viewable. Do not include private, confidential, or personally identifiable information. Private enterprise access controls are outside the project scope.
NFR-3	Reliability	Chart values should follow the connected dataset and workbook aggregations. Check that worksheets and dashboards load and display as intended before sharing. No formal reliability test is claimed.

NFR-4	Performance	The design should remain practical for the supplied dataset of approximately 27,468 records. Actual load times and performance benchmarks have not been measured; no numeric guarantee is specified.

Functional requirements define the capabilities the solution must provide. The requirements below reflect Disha's existing Tableau workbook and its three dashboard scenarios.

ID	Functional requirement	Requirement / acceptance intent	Priority
FR-1	Data Access & Integration	Connect the supplied global fuel-price CSV to Tableau Public. Use the fields date, country, region, income_level, subsidy_level, petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd, and tax_percentage. No separate database or live API is part of the current solution.	High
FR-2	Data Processing	Confirm field types and roles in Tableau; treat date as a time field and country, region, income level, and subsidy level as categories. Aggregate prices and tax measures for comparisons. Document cleaning or missing-value handling only if it was actually performed.	High
FR-3	Business Analysis	Compare average petrol, diesel, and LPG prices and average tax percentage. Support country rankings, regional comparisons, taxation-versus-price analysis, subsidy comparisons, income-level comparisons, and time trends alongside Brent crude prices. Treat associations as patterns, not proof of causation.	High
FR-4	Business Insights & Outputs	Provide three dashboards: (1) Fuel Taxation & Subsidy Policy; (2) Global Fuel Prices & Supply Chain Planning; and (3) Fuel Prices, Taxation & Economic Development. Include rankings, regional averages, tax/price views, subsidy and income-level comparisons, and historical trends.	High
FR-5	User Interaction & Analysis	Allow viewers to compare countries and regions, inspect top and bottom 10 countries by average petrol price, compare petrol and diesel prices, and use available Region legends and Income Level filters where included. Users explore through charts and filters rather than editing source data.	High
FR-6	Data Update & Reporting	Provide access to the published Tableau Public workbook. The current analysis covers the supplied dataset from January 2020 to April 2026 and does not claim automatic or real-time updates. The project owner must update the source, refresh the workbook, and republish when required.	Medium

2. Non-functional Requirements

Non-functional requirements describe quality expectations and constraints. Where no formal measurement has been performed, the document states the intended requirement rather than claiming a tested result.

ID	Quality attribute	Requirement / constraint
NFR-5	Availability	Viewing depends on Tableau Public, the published workbook, and an internet connection. No uptime guarantee is claimed.
NFR-6	Scalability	The current workbook is designed for the supplied dataset and three scenarios. Larger datasets or more frequent updates may require optimization, redesign, or another architecture; load testing has not been performed.

Scope and accuracy note: These requirements describe the current CSV-to-Tableau learning project. They do not imply a separate database, automated live ingestion, formal usability testing, load testing, or causal proof from the visualized relationships.

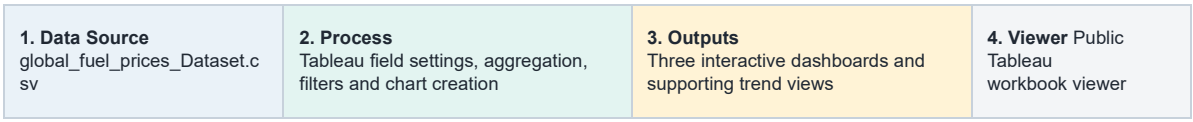
3. Technology Stack (Architecture & Stack)

The technology stack identifies the tools used to load, analyze, visualize, and publish the data. It is based on the documented project workflow rather than adding technologies that are not part of the implementation.

#	Component	Technology	Justification
1	Programming / query logic	Tableau calculated fields and built-in aggregations	Support measures such as average petrol price and average tax percentage. No separate Python or SQL pipeline is documented.
2	Data source / storage	CSV file: global_fuel_prices_Dataset.csv	The supplied CSV is the structured source. No separate warehouse or database such as Snowflake or MySQL is used.
3	Data ingestion	Tableau Public connection to the CSV	Loads the file as a data source for use in worksheets and dashboards.
4	Data transformation / business logic	Tableau data-source settings, field roles/types, filters, and aggregations	Analysis is performed in Tableau; no separate ETL pipeline or external cleaning process is claimed.
5	Visualization / analytics	Tableau Public 2026.3 worksheets and dashboards	Three dashboards cover taxation/subsidy policy, supply-chain price comparisons, and fuel prices by economic-development indicators.
6	User interface	Published Tableau Public workbook	Viewers interact with dashboards, chart selections, legends, and available filters such as income level.

4. Data Flow Diagram

The project-level data flow is a simple CSV-to-Tableau workflow: the supplied CSV is connected to Tableau, the data is aggregated and visualized in worksheets, the worksheets are arranged into three dashboards, and a public viewer explores the published workbook.



Flow: CSV source → Tableau data connection and analysis → worksheets / dashboards → public viewer. The diagram represents the implemented workflow at a high level; it does not imply a database, external API, automated refresh, or account system.

5. User Stories

User stories express the capabilities from the perspective of a dashboard viewer or the project owner. Acceptance criteria are observable checks for the published workbook or documented maintenance workflow.

User type	Epic / capability	ID	User story / task	Acceptance criteria	Priority	Release
External viewer	Explore dashboard insights	US-1	As a viewer, I want to open the published Tableau workbook so that I can explore global fuel-price and taxation information.	Workbook opens and the three scenario dashboards are accessible.	High	Release 1
External viewer	Compare taxation and subsidy policy	US-2	As an energy-policy viewer, I want to compare average fuel taxes and petrol prices by country or region so I can inspect patterns.	Dashboard 1 presents taxation, regional price, tax-versus-price, and subsidy-level views.	High	Release 1
External viewer	Review country price rankings	US-3	As a supply-chain viewer, I want to see the highest and lowest average petrol-price countries so I can compare potential cost differences.	Dashboard 2 shows top-10 and bottom-10 rankings of fuel prices of countries around world and petrol/diesel comparison.	High	Release 1
External viewer	Review historical price movement	US-4	As a supply-chain viewer, I want to review fuel-price trends alongside Brent crude prices to examine how the series move over time.	A historical fuel-price and Brent crude visualization is available; it supports comparison, not causal proof.	High	Release 1
External viewer	Explore income-level patterns	US-5	As an economics viewer, I want to compare average petrol prices and taxes across income levels to examine group-level differences.	Dashboard 3 includes average petrol price and average tax percentage by income level.	High	Release 1

External viewer	Filter analytical views	US-6	As an economics viewer, I want to use the income-level filter on relevant charts to focus on a selected group.	The filter is available on relevant Dashboard 3 views and updates them as configured.	Medium	Release 1
External viewer	Interpret categories consistently	US-7	As a viewer, I want clear category labels and legends so I can interpret regions, subsidy levels, and groupings correctly.	Relevant charts display titles, labels, and legends.	Medium	Release 1
Project owner / analyst	Maintain source data and workbook	US-8	As the project owner, I want to update the CSV source and refresh the workbook when needed so the analysis reflects a revised dataset.	Owner updates the source in Tableau and republishes; no automatic live feed is assumed.	Medium	Future maintenance
Project owner / analyst	Review data field definitions	US-9	As the project owner, I want to check source fields and data types so dates, prices, tax percentages, and categories are used appropriately.	Source fields include date, country, region, income_level, subsidy_level, petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd, and tax_percentage.	High	Release 1
External viewer	Access workbook without app-specific login	US-10	As a viewer, I want to view the publicly published workbook without a project-specific login flow so I can access dashboards easily.	Workbook is publicly viewable through Tableau Public; no private account or registration feature is claimed.	Medium	Release 1

Release labels are planning labels, not evidence of formal software sprints or a production release process.

6. Customer Journey Map

This journey map describes a plausible viewer's experience of accessing and interpreting the published Tableau dashboards. It is a design-stage hypothesis based on the workbook and intended analytical tasks, not a result of recorded customer interviews or formal usability testing.

Journey dimension	Stage 1 — Access & Orient	Stage 2 — Explore & Compare	Stage 3 — Interpret & Decide
Actions	Open the Tableau Story/workbook, read the title and identify the relevant question.	Select a dashboard, compare charts, use available filters and legends, and compare countries, regions, or groups.	Review rankings and trend patterns; compare price, tax, subsidy, income-level, or crude-oil views; note limitations.
Touchpoints	Tableau Public landing view, Story entry point, dashboard titles and chart labels.	Three dashboards, chart tooltips, legends, and available filters such as region or income level.	Rankings, time-series charts, tax/price comparisons, and observations for a presentation or report.
Thoughts	What does this workbook cover? What period and measures are shown? Which dashboard fits my question?	Which countries or regions differ? How do prices vary by subsidy or income level? Are measures comparable?	What patterns are visible? Are these associations or proven causes? What context is still needed?
Feelings	Curious; may need brief orientation to the dashboard structure.	Engaged when comparisons are clear; may be uncertain if filters or measures are unfamiliar.	More informed if patterns are clear; cautious about drawing conclusions from descriptive data.
Ownership	Project creator maintains the published workbook and dataset description.	Viewer chooses filters and interprets charts; owner maintains labels and chart configuration.	Viewer assesses patterns; project creator documents scope, definitions, and analytical limitations.
Improvement opportunities	Add a short "how to use" note; make date range and units easy to find; explain key measures.	Keep filter labels consistent, use clear titles/tooltips, and avoid overcrowding charts.	State that trends show association rather than causation; document dataset coverage and refresh process; validate interpretations with external context.

Project links

Tableau Story:
https://public.tableau.com/app/profile/harshvardhan.gupta1425/viz/Global_Fuel_Prices_and_Taxation_Trends_17932843176370/Story1

Project website: https://fuel-prices-taxation-dashboard.onrender.com/

GitHub repository: https://github.com/Harshvardhan-30/global-fuel-prices-taxation-analysis/tree/main

Accuracy note: The main Tableau link is the Story entry point, not a URL for an individual chart. The current analysis uses the supplied CSV and its documented coverage (January 2020–April 2026). The workbook is public and does not claim login, registration, automated data ingestion, or real-time updates.